MILITARY STANDARD

MIL-STD-1320-1?9 (NAVY)

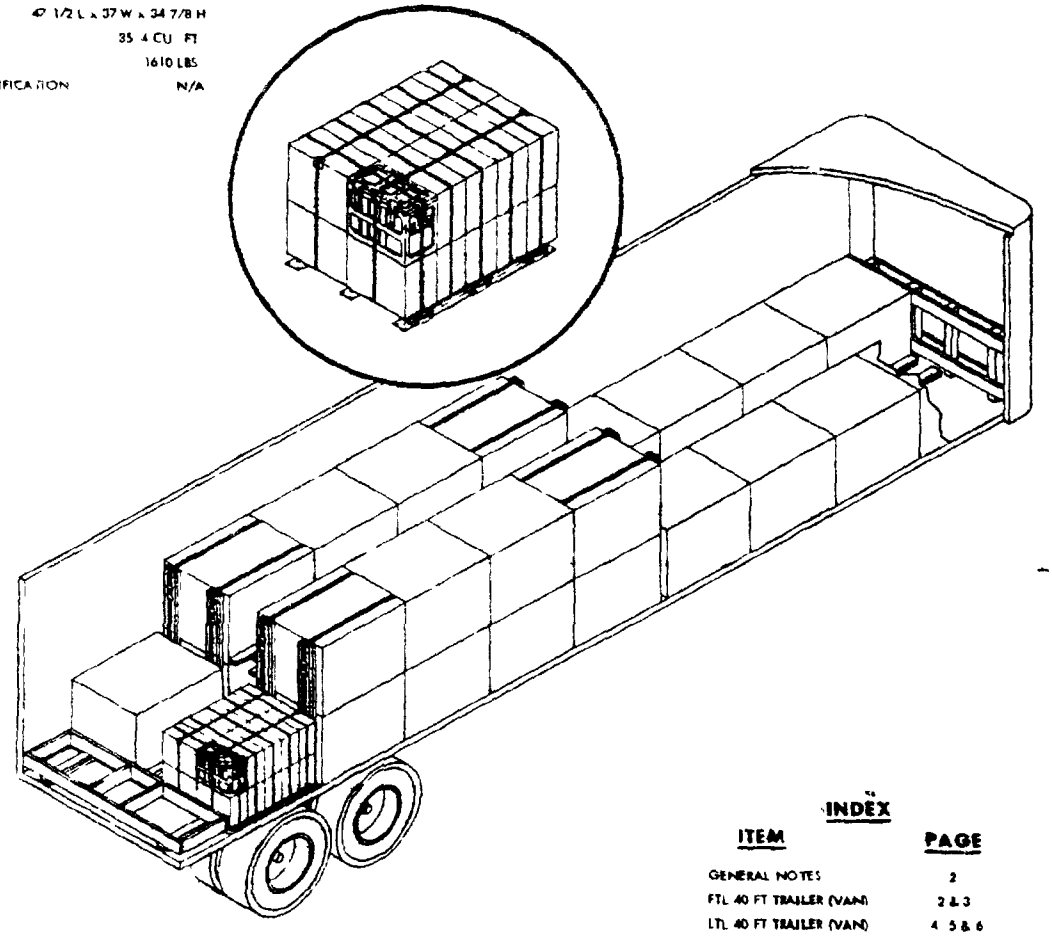
TRUCKLOADING OF HAZARDOUS MATERIALS

20 AUGUST 1973

WING ASSEMBLIES AND TAIL FINS
GUIDED MISSILE AGM-45A (SHRIKE)
IN CONTAINER CNU-171/E
FLEET ISSUE UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-54/248
DIMENSIONS	47 1/2 L x 37 W x 34 7/8 H
CUBE	35.4 CU FT
WEIGHT	1610 LBS
DOT HAZARD CLASSIFICATION	N/A



INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL 40 FT TRAILER (VAN)	2 & 3
LTL 40 FT TRAILER (VAN)	4 5 & 6
UNITIZING PROCEDURE	6

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINER-LOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

SIGNATURE AIRSYSCOM BY DIRECTION DATE

ORIGINATOR Marie McBean 9/14/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

PAGE 1 OF 7

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GENERAL NOTES

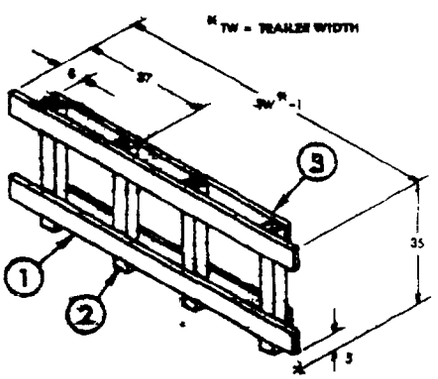
1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING WING ASSEMBLIES AND TAIL FINS FOR THE GUIDED MISSILE AGM-45A (SHRIKE) WHEN PACKED IN CONTAINER CNU-171/E AND PALLETIZED MB-54/24B.
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT VAN TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. THE LENGTH OF THE STRUT, PIECE 5, SHALL BE SUFFICIENT TO CAUSE THE REAR BLOCKING ASSEMBLY, DETAIL B, TO BEAR AGAINST THE UNIT LOADS AND THE TRAILER DOORS WHEN THE TRAILER DOORS ARE CLOSED.

WARNING

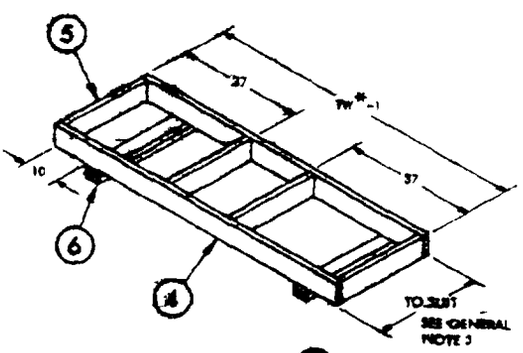
DO NOT USE TRAILERS WITH ROLL UP REAR DOORS WHEN BLOCKING AND BRACING BY THIS METHOD

4. SWAY BRACE, DETAIL C, IS ASSEMBLED IN PLACE. INSERT SUPPORTS, PIECE 7, IN PLACE AND AGAINST INBOARD SIDE OF PALLET STRINGERS. POSITION STRINGERS, PIECE 8 AGAINST PALLET STRINGERS AND NAIL TO PIECE 7 WITH THREE 12d NAILS PER JOINT.
5. TO PREVENT LONGITUDINAL MOVEMENT OF UNIT LOADS WHEN THERE IS A SECOND LAYER, THE UNIT LOADS SHALL BE UNITIZED (SEE UNITIZING DETAIL, PAGE 61) WHERE THE LOAD CHANGES FROM TWO LAYERS HIGH TO ONE LAYER HIGH. THE STIFFENERS SHALL BE POSITIONED TOWARDS THE UNIT LOADS THAT ARE ONE HIGH. IF BOTH ENDS OF A STACKED LOAD ARE AGAINST ONE HIGH UNIT LOADS, STIFFENERS SHALL BE USED ON BOTH ENDS OF THE STACKED TWO HIGH LOADS.
6. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA AND CLOSE AND SEAL TRAILER DOORS.
8. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM-L-751; NAILS, FF-N-105, TYPE II STYLE 10, COMMON BRIGHT; STRAPPING, QQ-S-781, TYPE I CLASS A.
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

FTL 40 FT TRAILER (VAN)

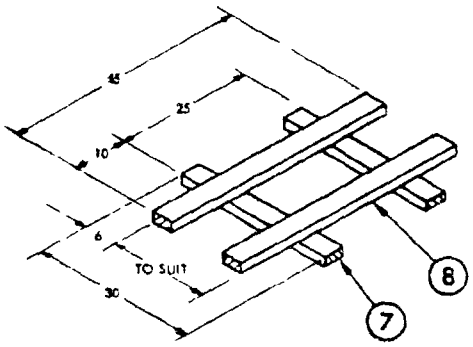


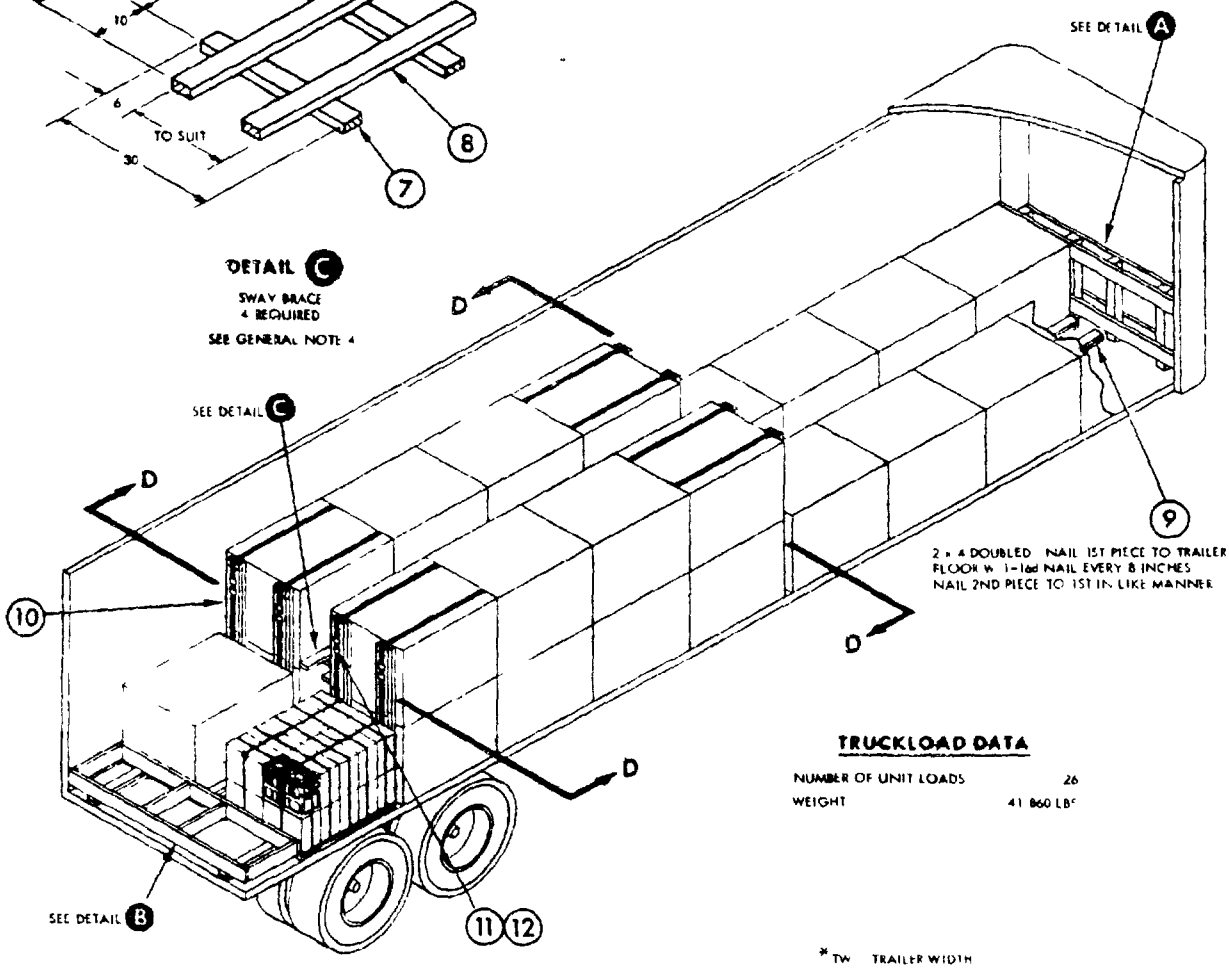
DETAIL A
FRONT BULKHEAD
1 REQUIRED



DETAIL B
REAR BLOCKING ASSEMBLY
1 REQUIRED

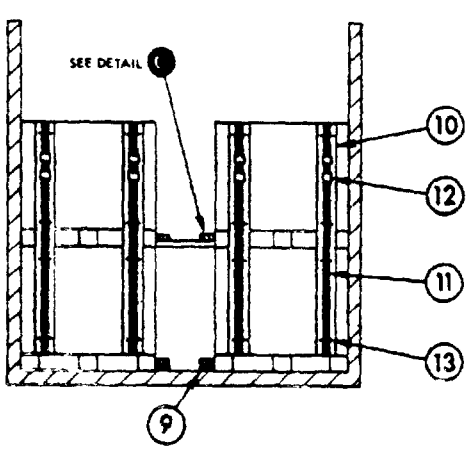
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TRUCKLOAD DATA

NUMBER OF UNIT LOADS	26
WEIGHT	41 860 LBS



SECTION D-D

* TW TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
13	STAPLE	FOR 1 1/4 STRAP	32			-
12	SEAL	FOR 1 1/4 STRAP	16	-	-	-
11	STRAP	1 1/4 x .035 x 21 FT	8			-
10	STIFFENER	2 x 6 x 64 3/4	16	DBLD & LAMINATED	5	10d
9	SLEEPER	2 x 4 x 45	36	SEE FIELD NOTE		
8	STRINGER	2 x 4 x 45	8	7	3 PER JOINT	12d
7	SUPPORT	2 x 4 x 30	8	SEE 8	-	-
6	SUPPORT	2 x 4 TO SUIT	4	4 6	2 PER JOINT 4	12d 10d
5	STRUT	2 x 6 TO SUIT	4	SEE 4		
4	CROSSMEMBER	2 x 6 x TW*-1	2	5	3 PER JOINT	16d
3	FWD HORIZONTAL	2 x 6 TO SUIT	2	2	3 PER JOINT	12d
2	VERTICAL	4 x 4 x 35	4	SEE 1 & 3	-	-
1	AFT HORIZONTAL	2 x 6 x TW*-1	2	2	3 PER JOINT	12d

LIST OF MATERIALS AND NAILING DATA

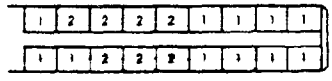
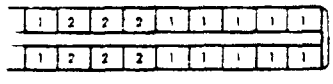
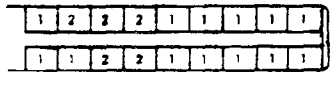
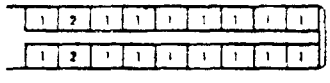
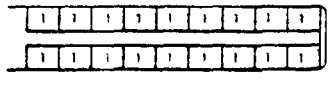
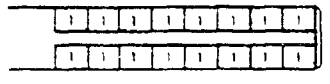
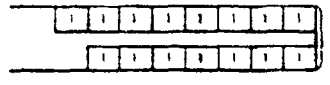
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LTL 40 FT TRAILER (VAN)

1 WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE I

2 WHEN SHIPPING 18 TO 25 UNIT LOADS FOLLOW THE BASIC PRINCIPLE SHOWN FOR THE LTL ON PAGE 5 WHEN SHIPPING LESS THAN 18 UNIT LOADS FOLLOW THE PRINCIPLES SHOWN ON PAGE 6

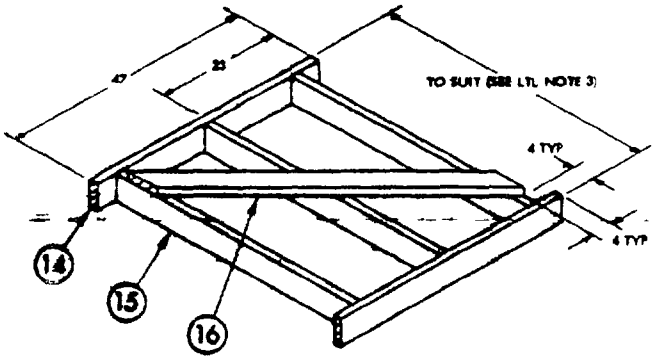
TABLE I

NO OF UNIT LOADS	LOAD PATTERN* FWD →
25	
24	
23	
22	
21	
20	
19	
18	
17	
16	
15	
14 OR LESS	POSITION UNIT LOADS IN TRAILER USING A "DOUBLE ROW" PATTERN BLOCK AND BRACE AS SHOWN ON PAGE 6

*LOAD PATTERN SHOWS FLOOR PLAN FOR TRAILER NUMBER IN BLOCKS INDICATES NUMBER OF UNIT LOADS IN LAYER.

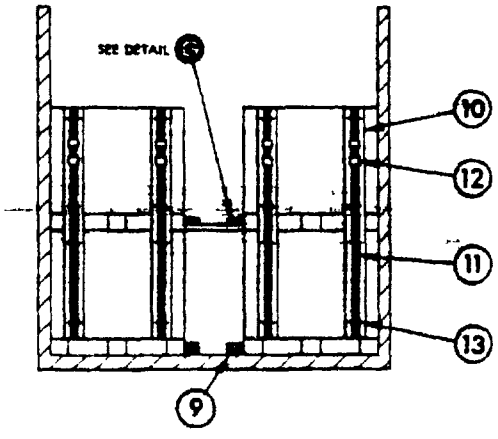
3 THE LENGTH OF THE STRUTS PIECE 15 SHALL BE SUFFICIENT TO CAUSE THE SWAY BRACE ASSEMBLY DETAIL E, TO BEAR AGAINST THE UNIT LOAD AND THE TRAILER SIDE WALL TWIST THE ASSEMBLY TO THE UNIT LOAD BENEATH (4 PLACES) USING 16 GAUGE SOFT ANNEALED IRON WIRE

4 THE LTL ON PAGE 5 IS FOR 21 UNIT LOADS AND THE LTL ON PAGE 6 IS FOR 5 UNIT LOADS THIS DOES NOT LIMIT LTL SHIPMENTS TO THIS QUANTITY ACTUAL SHIPPING REQUIREMENTS DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED THE LOAD PATTERN AND DUNNAGING SHALL BE IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THIS DOCUMENT AND THE GENERAL TRUCK-LOADING STANDARD, MIL-STD-1320 (NAVY)



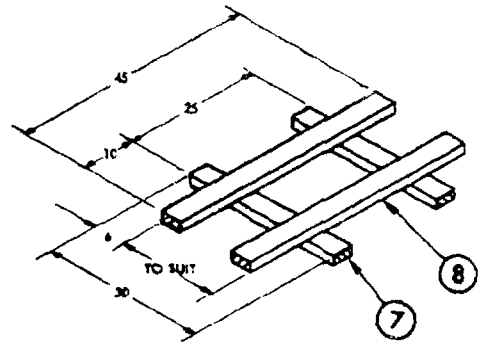
DETAIL F

SWAY BRACE ASSEMBLY
1 REQUIRED
SEE LTL NOTE 3



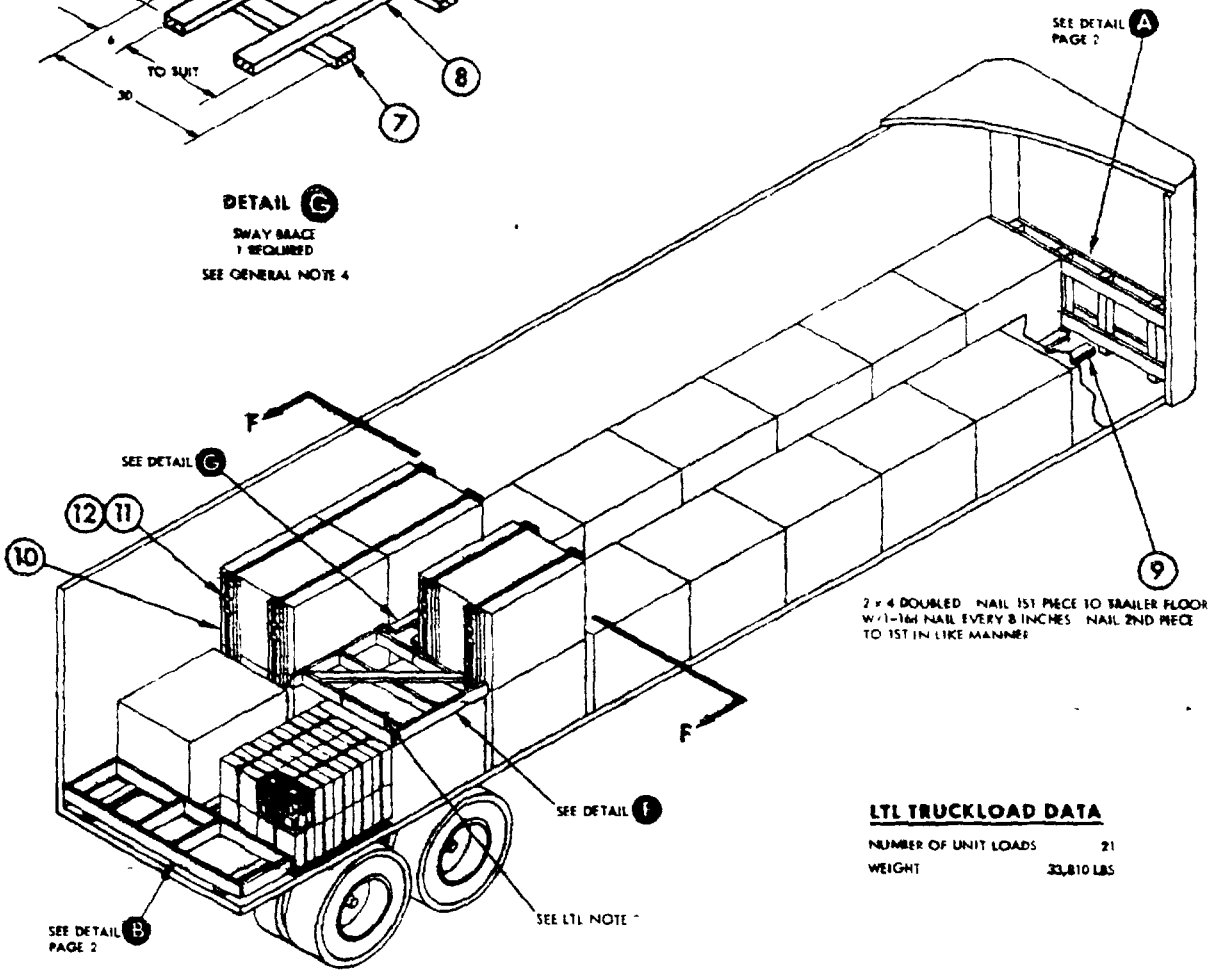
SECTION F-F

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DETAIL G

SWAY BRACE
1 REQUIRED
SEE GENERAL NOTE 4



2 x 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

LTL TRUCKLOAD DATA

NUMBER OF UNIT LOADS	21
WEIGHT	33,810 LBS

*TW = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
16	DIAGONAL	2 x 4 TO SUIT	1	15	2 PER JOINT	13d
15	STRUT	2 x 6 TO SUIT	3	SEE 14	-	-
14	LONGITUDINAL	2 x 6 x 47	2	15	9 PER JOINT	16d
13	STAPLE	FOR 1 1/4 STRAP	32	-	-	-
12	SEAL	FOR 1 1/4 STRAP	12	-	-	-
11	STRAP	1 1/4 x .035 x 21 FT	4	-		-
10	STIFFENER	2 x 6 x 64 3/4	16	BEND & LAMINATED	5	10d
9	SUBRAIL	2 x 4 x 48	26	SEE FIELD NOTE		

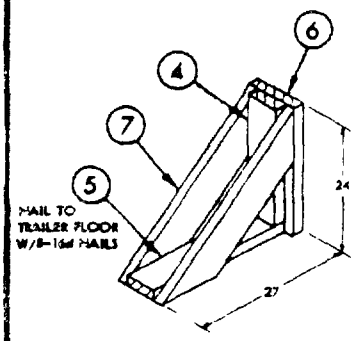
LIST OF MATERIALS AND NAILING DATA

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
8	STRINGER	2 x 4 x 48	2	7	3 PER JOINT	13d
7	SUPPORT	2 x 4 x 30	2	SEE 9		-
6	SUPPORT	2 x 4 TO SUIT	4	4	2 PER JOINT	13d
5	STRUT	2 x 6 TO SUIT	4	SEE 4	-	-
4	CROSSMEMBER	2 x 6 x TW*-1	2	8	3 PER JOINT	16d
3	FWD HORIZONTAL	2 x 6 TO SUIT	2	2	3 PER JOINT	13d
2	VERTICAL	4 x 4 x 35	4	SEE 1 & 3	3 PER JOINT	13d
1	AFT HORIZONTAL	2 x 6 x TW*-1	2	2	3 PER JOINT	13d

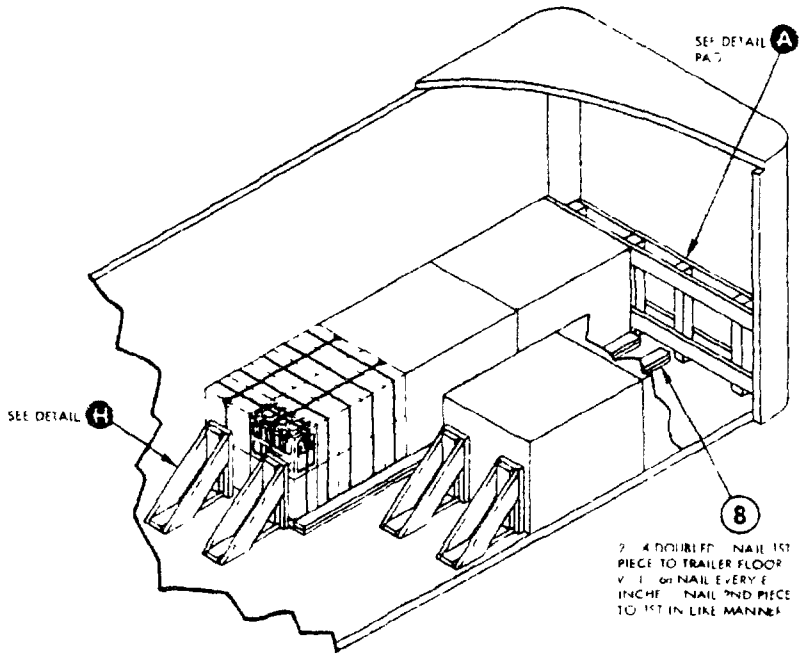
LIST OF MATERIALS AND NAILING DATA

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MIL-STD-1320-119 (NAVY)



DETAIL H
KNEE BRACE
4 REQUIRED



*Tw = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
	LIST OF MATERIALS AND NAILING DATA				NAILS	
8	SLEEPER	2 x 4 x 45	10	SEE FIELD NOTE		
7	BRACE DIAGONAL	2 x 6 TO SUIT	8	4 & 5	4 PER JOINT	12d
6	BRACE BACKUP	2 x 10 x 24	4	4	5	10d
5	BRACE CLEAT	2 x 6 x 24	4	4	3	16d
4	BRACE UPRIGHT	2 x 6 x 24	4	SEE 2 & 6		
3	FWD HORIZONTAL	2 x 6 TO SUIT	2	2	3 PER JOINT	12d
2	VERTICAL	4 x 4 x 95	4	SEE 1 & 3	-	-
1	AFT HORIZONTAL	2 x 6 x TW*-1	2	2	3 PER JOINT	12d

LTL TRUCKLOAD DATA

NUMBER OF UNIT LOADS	5
WEIGHT	8 050 LBS

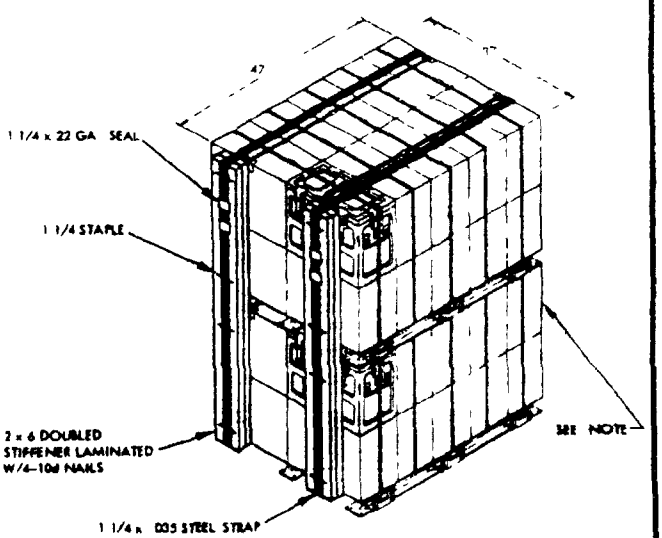
LTL 40 FT TRAILER (VAN)

UNITIZING PROCEDURE

1. WHEN REQUIRED BY THE FTL OR LTL REQUIREMENTS OF THIS MIL STD THE TWO HIGH STACKED UNIT LOADS SHALL BE STRAPPED AS SHOWN ABOVE
2. THE 2 x 6 DOUBLED STIFFENER SHALL EXTEND FROM THE TOP OF STACKED UNIT LOADS TO THE TOP OF THE PALLET OF THE BOTTOM UNIT LOAD. THE STIFFENER SHALL BE LAMINATED WITH FOUR 10d NAILS TO PREVENT SHIFTING
3. THE 1 1/4 INCH STEEL STRAPS POSITION AS SHOWN ENCIRCLE THE STACKED UNIT LOADS AND PASS UNDER THE DECK OF THE BOTTOM PALLET WITH THE STRAPS HOLDING THE STIFFENERS IN PLACE
4. TENSION STRAPS AND SEAL WITH TWO DOUBLED CRIMPED SEALS SECURE EACH STRAP TO THE STIFFENER WITH FOUR 1 1/4 INCH STAPLES

NOTE

WHEN LOADING A TRAILER WITH 19, 20, AND 21 UNIT LOADS THE SINGLE STACKED UNIT LOAD (NOT BACKED UP BY ANOTHER STACKED LOAD SHALL HAVE STIFFENERS AT BOTH ENDS OF THE STACKED LOADS. SEE GENERAL NOTE 5 PAGE 2



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REVIEW ACTIVITIES:
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - O
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